



Other Properties of Complex Numbers

1. Equality of Complex Numbers

Two complex numbers $a + jb$ and $c + jd$ are equal if and only if $a = c$ and $b = d$.

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2. Conjugate of Complex Numbers

The conjugate of the complex number $a + jb$ is $a - jb$.

3. Multiplicative Inverse (Reciprocal) of Complex Numbers

The reciprocal or the multiplicative inverse of the complex number $a + jb$ is defined as $1/(a + jb)$.

Matrices

Matrix is a rectangular array of real numbers arranged in m rows and n columns. The term "matrix" was introduced by the English mathematician **James Joseph Sylvester** (1814 – 1897) in 1850. The size of a matrix is determined by the number of rows and columns. The expression " $m \times n$ " is the **dimension** or **order** of the matrix. The first non-zero entry in a row of a matrix is known as the **leading entry** or the **leading element**.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

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where:

A = name/label of the matrix

m = number of rows

n = number of columns

$m \times n$ = size/dimension/order of the matrix

$a_{11}, a_{12}, \dots, a_{mn}$ = elements of the matrix
(first subscript always denotes the row and the second subscript always denotes the column)

If the matrix has only one column, it is called a **column matrix** and if it has only one row, it is called a **row matrix**.

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The matrix below is an example of a **column matrix**. It is also called a column vector. For this type of matrix, $n = 1$.

$$A = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$$

The matrix below is an example of a **row matrix**. It is also called a row vector. For this type of matrix, $m = 1$.

$$A = [a_{11} \quad a_{12} \quad \cdots \quad a_{1n}]$$

Usually, matrix names are denoted by capital boldface letters. Lowercase boldface letters are used to denote vectors.

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If the matrix has an equal value of m and n , then it is called a **square matrix**. The matrix below is an example of this type of matrix. It has a size of 3×3 .

$$A = \begin{bmatrix} 6 & 5 & 4 \\ 2 & 1 & 4 \\ 3 & -2 & -2 \end{bmatrix}$$

For square matrices, the diagonal from the upper left to the lower right is called the **principal diagonal** or **main diagonal** and all entries in the said diagonal are called as **diagonal entries**.

$$A = \begin{bmatrix} 6 & 5 & 4 \\ 2 & 1 & 4 \\ 3 & -2 & -2 \end{bmatrix}$$

In the matrix above, 6, 1 and 2 are the diagonal entries. The sum of all diagonal entries of a square matrix is called the **trace of the matrix**.

Types of Matrices

1. Square Matrix

This type of matrix has the same number of columns and rows.

$$A = \begin{bmatrix} 6 & 5 & 4 \\ 2 & 1 & 4 \\ 3 & -2 & -2 \end{bmatrix}$$

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2. Diagonal Matrix

This type of matrix is a square matrix that has values only in its diagonal, the rest of the elements are 0.

$$A = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

3. Scalar Matrix

This type of matrix is a diagonal matrix whose diagonal entries are all equal.

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

4. Identity or Unit Matrix

This type of matrix is a scalar matrix whose non-zero elements are all equal to 1.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5. Zero or Null Matrix

This type of matrix is a any matrix that does not contain any non-zero element.

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

6. Triangular Matrix

Lower Triangular Matrix is a diagonal matrix whose entries above the main diagonal are all zero.

$$A = \begin{bmatrix} 6 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -2 & -2 \end{bmatrix}$$

Upper Triangular Matrix is a diagonal matrix whose entries below the main diagonal are all zero.

$$A = \begin{bmatrix} 6 & 5 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & -2 \end{bmatrix}$$

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